Odd-characteristic lift-dense lines: orbit parity and descent

Odd-characteristic lift-dense lines: the orbit-parity criterion holds, and descent lines lose their lifted half

2026-11-06. This is the second piece checking the claims that reached this repository from another session as an unverified assertion. The first piece was `lift_dense_torsion_pairing_20261106`.

The protocol was frozen before any data, at `89611afe58` (`PROTOCOL.md`; deviations under “Amendments”). The instrument is `orbit/`, std-only Rust written fresh for this check. The run is `reports/run-20261106T0306Z/`, at commit `eedd7b0a6d`. Report: `REPORT.pdf`.

Every statement below is an unverified assertion of this session [pcc:unverified]. The claim record is at the end.

The claims checked

E is $y^2 = f(x)$, a cubic over K = F_{q^n} with q = p odd. A line is $x_0 + u \cdot F_q$. The shifted cubic $F(t) = f(x_0 + ut)/u^3$ has roots $t_i = (e_i - x_0)/u$.

1. **Criterion.** A line is all-lift or all-twist exactly when, for each q-Frobenius orbit O of the roots of F, n·r_O/s_O is even.
2. **Descent case.** Half the lifted points sit on the twist and are dead columns for the target.

Answer

- **The criterion holds in both directions at the scope tested** (q ∈ {13, 101}, n ∈ {2, 3, 4}).
  - All 1,287 lines that satisfy it are constant, with 0 exceptions.
  - Where the Weil bound leaves no room for chance, (101,2) and (101,3), none of 3,520 lines that fail it is constant.
  - At q = 13, one line out of 1,600 at n = 3 fails the criterion and is still constant: all 13 values are nonsquares. That is the chance constancy the

Weil bound allows at small q, as registered.
    - The multiplicity identity behind the criterion (n·r/s is an integer and the orbit masses sum to 3n) held on every line.
- **Descent lines.** These are lines whose root set is σ^{n/2}-stable.
    - They exist on curves with j(E) ∈ K_0 = F_{q^{n/2}}, which are subfield curves up to isomorphism: 32 built curves, all with j ∈ K_0.
    - Each of the 1,280 built lines satisfies the criterion and has F ∈ K_0[t].
    - Each built line is all-lift exactly when χ_K(u^3) = 1, so on one curve the built lines are either all all-lift or all all-twist. There were 0 exceptions.
- **“Half dead” holds, but “predicted which with 0 mismatches” fails as registered.**
    - On all-lift descent lines, the pooled dead fraction is 0.505–0.615 per cell, inside the registered [0.35, 0.65].
    - The rule “dead iff the point lies on the side of #C(K_0)·#C′(K_0) that does not contain ℓ” mispredicts 405 of 27,880 lifted abscissae. Every one is the same kind: the point is observed dead but predicted alive. No predicted-dead point was alive.
    - So the twist side is always dead. The misses are points on the ℓ-side whose order is prime to ℓ (the cofactor part), which the rule ignores.
    - At (101,4), a uniform estimate of how many such points to expect gives 68, and 70 were observed. At n = 2 the estimate is too high (224 and 333 against 80 and 240). There the line is all of K_0, so a uniform estimate does not fit. That residual is unexplained.
- **What it means as a lever.** All criterion-satisfying lines seen here were descent lines; no other mechanism appeared.
    - All 1,287 criterion lines lie on curves with j ∈ K_0:
        - 1,280 are built descent lines;
        - 4 are random lines on descent curves;
        - 3 are random lines on the one generic curve whose j happens to lie in F_{13}, at (13,2).
    - Every random line on the other 31 generic curves in even-n cells (j ∉ K_0) failed the criterion.
    - The descent case lives only on subfield-type curves (j ∈ K_0), where the lifted half on the twist side is dead for the target. So it is no lever for the target subgroup.
    - That matches the n = 2 subfield finding of the first piece in characteristic 2.
    - The conjugate-pair family, which the source says gives useful columns on generic curves with full 2-torsion, is the next piece. It needs constructed lines, because random lines never hit it here.

Registered verdicts (`orbit summarize`)

rule	verdict	detail
O1 criterion ⇒ constant	**PASS**	1,287 criterion lines, 0 not constant
O2 ⇐ where Weil decides, (101,2) and (101,3)	**PASS**	3,520 lines fail the criterion, 0 constant
O2′ ⇐ elsewhere (report only)	—	1 chance-constant line at (13,3), 0 at (13,2), (13,4) and (101,4)
O3 descent lines	**PASS**	1,280 built lines: criterion, σ^{n/2}-stable, F ∈ K_0[t], all-lift iff χ_K(u^3) = 1; 0 exceptions
O4 dead set exact, fraction in [0.35, 0.65]	**FAIL**	520 decidable lines, 405 mismatches (all observed dead /

rule	verdict	detail
		predicted alive); fractions 0.505–0.615 (that part passes)
O5 multiplicity identity	**PASS**	0 exceptions on 12,160 lines
A/A	**PASS**	two runs give byte-identical `rows.jsonl` (sha256 `273431a9…`)

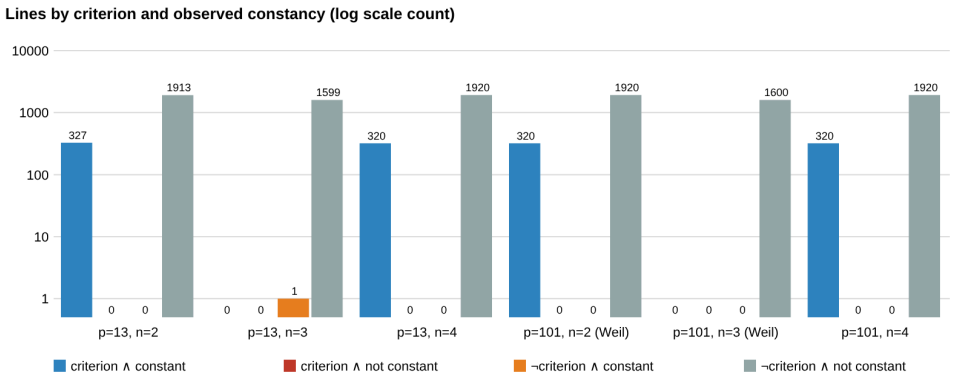
Curve checks.

- #E(K) = #$C_0(K_0)\cdot\#C_0'(K_0)$ (or its K-twist) passed [#E]P = O on 8 points of each of the 32 descent curves.
- ℓ divides both sides on 3 curves, which O4 excludes as registered.
- On 18 of the 32 descent curves α is a nonsquare, so their built lines are all-twist and do not enter O4.

Per cell

p	n	Weil decides	lines	criterion true	constant	criterion false, constant
13	2	no	2240	327	327	0
13	3	no	1600	0	1	1
13	4	no	2240	320	320	0
101	2	yes	2240	320	320	0
101	3	yes	1600	0	0	0
101	4	no	2240	320	320	0

p	n	decidable lines	lifted x	dead observed	dead predicted	mismatches	dead fraction	predicted dead, alive
13	2	160	2080	1280	1200	80	0.615	0
13	4	120	1560	840	825	15	0.538	0
101	2	120	12120	7160	6920	240	0.591	0
101	4	120	12120	6116	6046	70	0.505	0



lines by criterion and constancy

The full tables are in `SUMMARY.md`.

Not covered

- The conjugate-pair family needs n even, n ≥ 4, and full 2-torsion over K. The source says a generic such curve over F_{q^4} has 6 such lines, found from

one F_q-linear equation. It is not checked here.

- “Odd n has no lever” is consistent with these data: no n = 3 line met the criterion apart from the one chance-constant line. But that rests only on random lines, not on a search.
- The 2^m yield gain is not measured. No ECDLP cost was measured, so there is no leaderboard, benchmark or rho row.

Sources checked

The `research-kb` server and CLI are not available in this container. A grep of the three checkouts for odd-characteristic all-lift lines found nothing. The char-2 trace version is `lift_dense_factor_base_20261027`.

Reproduce

```
        cd orbit && cargo build --release && cargo test --release &&
cd ..
        ./orbit/target/release/orbit run reports/run-X          # ~75
s, 4 threads
        ./orbit/target/release/orbit summarize reports/run-X
        ./orbit/target/release/orbit validate reports/run-X
claimed.json
```

Claim record

claim	standing	content
`PCC-20261106-2ae3b9`	not covered (unsigned)	O1–O5 counts and the dead-point tallies (`validator-report.json`, headline values in `claimed.json`)